



Dynamics of leaf litter decomposition and nutrient release of selected tree species in Central Nepal Himalaya

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Abstract

Litter decomposition is a fundamental ecological process that regulates nutrient cycling and soil fertility in forest ecosystem. This study compared the litter decomposition rates and nutrient dynamics of dominant tree species at different elevations in the Gaurishankar Conservation Area of Nepal. Leaf litter decomposition was monitored using the litter bag method under natural field conditions for one year. Decomposition rates varied significantly among species, with *Pinus roxburghii* exhibiting the most rapid decomposition (89% mass loss), whereas *Pinus patula* had the slowest decomposition rate (33% mass loss). The Himalayan deciduous species, *Myrica esculenta*, decomposed more rapidly (52%) than the evergreen species such as *Quercus lanata* (46%) indicating the influence of functional traits on leaf litter decomposition. Overall, the decomposition rates followed the order: *P. roxburghii* > *Pinus wallichiana* > *M. esculenta* > *Q. lanata* > *Rhododendron arboreum* > *P. patula*. Similarly, macro-nutrient release from the decomposing litter followed the sequence phosphorus > potassium > carbon > nitrogen, indicating relatively strong nitrogen retention in litter. Elevation had a strong negative correlation with litter mass loss, indicating decreased the rate of decomposition as elevation increased. The estimated time required for 95% decomposition (t_{95}) ranged from 2.26 and 12.5 years for *P. roxburghii* and *P. patula*, respectively, demonstrating the role of elevation in nutrient retention. These findings, therefore, suggest that a combination of factors such as elevation, litter chemistry, and species-specific traits play a key role in nutrient cycling and soil fertility in the Himalayan forests.

Keywords: C:N ratio, decay constant, elevational gradient, Gaurishankar Conservation Area, nutrient dynamics

Introduction

Litter in functional plant ecology refers to the dead plant matter that separates from living plants and accumulates on the soil surface (Anderson and Ingram 1993) which plays an important role in forest ecosystem as a source of primary organic matter (OM). Litter decomposition is a major process in nutrient cycling affecting soil productivity (Bhattarai and Mandal 2018) and in shaping plant populations, either inhibiting or promoting growth (Baltzinger et al. 2012). In temperate forests dense litter layers suppress ground vegetation, while in, tropical forests, rapid decomposition fosters species diversity (Gautam and Mandal 2017). The pattern of litterfall varies according to the forest type. Evergreen forests experience continuous litterfall, whereas litterfall in deciduous forests peak in autumn (Bo-

hara et al. 2020). Seasonal litterfall patterns further influence ecosystem dynamics (Zhang et al. 2018). The litter decomposition process is regulated by environmental factors, litter quality, microbial activity, and soil nutrients (Berger et al. 2010; Berger and Berger 2012, 2014). The decomposition process fragments, leaches, and with microbial breakdown releases plant-available nutrients (Swift et al. 1979; Vitousek et al. 1994). Decomposition rates are also dependent on litter chemistry, including carbon-to-nitrogen (C:N) ratio, lignin, cellulose, and polyphenol content (Silveira et al. 2011). Nitrogen (N)-rich litter, of *Alnus nepalensis*, decomposes more rapidly than N-poor pine needles (Bohara et al. 2020). Further, rapid decomposition enhances nutrient release, whereas slower decomposition promotes the accumulation of forest-floor OM (Isaac and Achuthan Nair 2005; Kim 2012).



Furthermore, elevation gradients, climate, soil chemistry, and plant traits impact the decomposition process (Berger et al. 2015). High-elevation forests may function as carbon (C) sinks, stabilizing soil C and contributing to atmospheric sequestration (Bangroo et al. 2017). Nutrient cycling, particularly C, N, phosphorus (P), and potassium (K), are driven by litter decomposition and soil interactions, which are considered vital for forest productivity and ecosystem health (Hauer and Lamberti 2006; Palm and Sanchez 1990).

Use of forest litter for animal bedding and converting it into farm manure is a common practice in subsistence farming in the Gaurishankar Conservation Area (GCA). However, local farmers lack knowledge of litter types and decomposition processes, limiting their benefits from the ecosystem. Understanding decomposition rates and nutrient release dynamics can improve soil and litter quality assessments, benefiting farmland productivity for mountain communities. Despite the significance of litter in nutrient dynamics, research on litter decomposition and nutrient cycling in Nepal's middle mountain region remains limited. Therefore, the objectives of this study were to: i) evaluate in-situ litter decomposition rates of common tree species, ii) assess nutrient release dynamics of carbon, nitrogen, phosphorous, potassium, and iii) assess the relationship between decomposition rates and environmental factors such as elevation, temperature, and soil moisture. It is anticipated that these findings will offer baseline data to generalize the nutrient release pattern of some Himalayan tree species for future research.

Materials and Methods

Study area

The GCA is managed by the National Trust for Nature Conservation (NTNC), and it spreads across Sindhupalchok, Dolakha, and Ramechhap districts, in the Central Nepal Himalayas, covering 2,179 km² area. The area spans within an elevational range from 968 m to 7,181 m, encompassing subtropical to alpine climatic zones with diverse vegetation types, including *Juniperus*, *Pinus* spp., *Quercus* spp., *Rhododendron* spp., *Schima-Castanopsis*, and *A. nepalensis* forests. The land cover consists of 34.98% forest, 9.8% shrubland, 8.61% grassland and pasture land supporting 565 species of wild flora (Department of National Parks and Wildlife Conservation 2026).

The study was conducted in Bigu Rural Municipality, Dolakha District, that has a predominantly agrarian economy, where locals engage in subsistence terraced farming and livestock rearing. The major crops grown in the area include potatoes, wheat, barley, millet, and buckwheat at higher elevations, while rice is grown in lower, monsoon-dependent areas.

Six study sites, spanning elevations from 1,010 m to 2,500 m, were selected across Singati, Laduk, Bulung, and Bulungkhani settlements (Fig. 1) of ward number 1 to 4 of Bigu Rural Municipality. Site selection was based on dominating tree species along the elevational gradient to study ecological variations, comprehensively.

The climate of the study area ranged from sub-tropical at Singati (950 m) to upper cool temperate at Bulungkhani (2,500 m). Singati has a typical monsoon climate with hot and humid summer temperatures reaching up to 32°C, while winter is cooler, whereas Bulungkhani has cooler

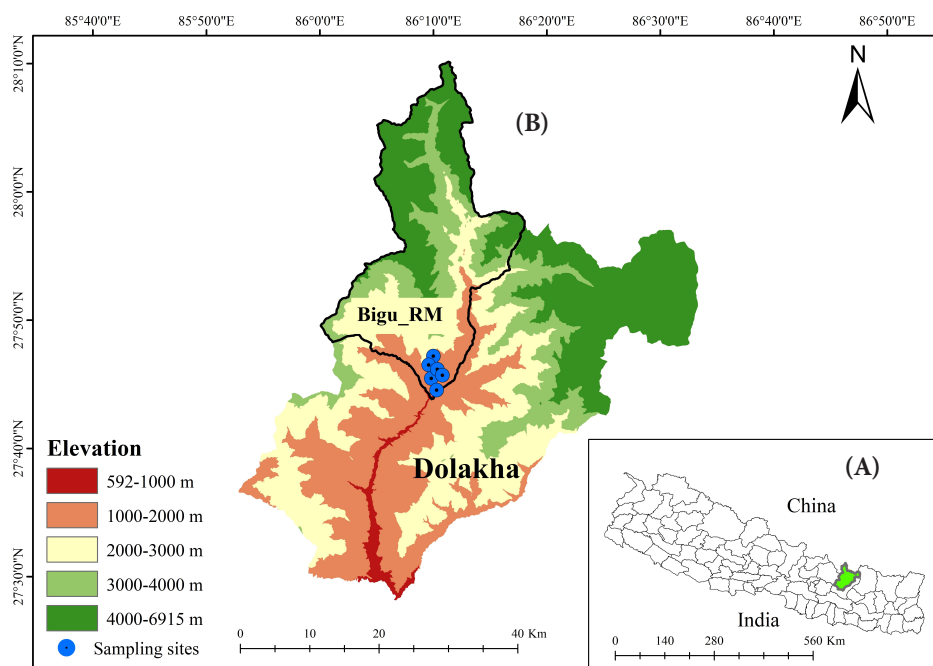


Fig. 1 Study area map showing Dolakha district (green) of Nepal (A) and sampling sites in Bigu rural municipality of Dolakha (B).

temperatures with frosty winters and receives snow during January and February, dropping the temperature to sub zero.

The tree species were purposively selected, depending on their dominance in the forest stands along the elevational gradient. There was a natural separation between *Pinus roxburghii* forests at lower elevations and other selected tree species at higher elevations (Table 1), often demarcated by settlements and agricultural fields, highlighting the ecological transitions and the dynamic nature of the landscape.

Litter decomposition study

The study followed established protocols for litter collection, litter bag deployment, and soil sampling as described by Berger et al. (2015) and Bohara et al. (2020). Freshly fallen litter were collected manually from 1 m² plots in selected stands, dried at room temperature until a constant weight was achieved, and stored in labeled zip-lock bags till further analysis. Litter bags, prepared from 1 mm polyethylene mesh (10 cm × 10 cm) and secured with clips, were filled with 3 g of dried litter. A total of 96 litter bags (16 per species, with four replicates per season) were deployed at six sites in June 2022, and the last replicates of litter bags were retrieved in June 2023.

For each site, 16 litter bags were tied with strings (4 replicates in one string) and secured with metal pegs on the ground surface. Litter bags containing the sampled litter were removed carefully in June 2022 leaving no trace of litter bags before monsoon. Photographs along with GPS coordinates of each site were recorded. Four litter bags per species were retrieved at 3-month intervals over one year and the last set of replicates of the litter bags were retrieved in May 2023. The retrieved bags were gently cleaned with a brush to remove ingrown roots and soil, dried to a constant weight at temperature (25°C), and weighed using a precise weighing machine (PGB220; Wensar, Chennai, India). The residual mass was stored in labeled bags for nutrient analysis. Litter samples were oven dried at 60°C for 24 hours prior to analysis. The oven dried samples were ground to a fine powder and sealed in zipper bags for further lab analysis. The samples were chemically analyzed in the Agriculture Technology Center, Kathmandu.

Litter N was determined using the Kjeldahl Digestion-distillation method (Kjeldahl 1883), phosphorus (P₂O₅)

with the Vanadomolybdate Spectrophotometric method, and potassium (K₂O) with Flame Photometry (Lundegardh 1934). OM was assessed using the Walkley-Black method (Walkley and Black 1934).

Data management and analysis

To evaluate the loss of litter mass over time (%), the protocols of Bragazza and Iacumin (2009) and Olson's (1963) Decay Model were employed:

$$K = -\ln(X_t/X_0)/t$$

Here, X_t is the final litter weight (g) after collection, X_0 is the initial litter weight (g) at placement, $\ln$ is the natural logarithm base, K is the decay constant, and t is the time interval in years.

Nutrient release from litter (%) was calculated using the formula from Guo and Sims (1999):

$$R = \{(X_0C_0 - X_tC_t) / X_0C_0\} \times 100$$

Where 'R' is the release of nutrients (%), ' C_0 ' is the 'Initial Nutrient Concentration (mg g⁻¹)', and ' C_t ' is the 'final concentration of nutrient' in the litter.

Data variations and correlations were analyzed. Differences in litter weight reduction across species and elevational gradients were analyzed using a one-way analysis of variance (ANOVA), after verifying data normality and identifying outliers. When ANOVA indicated significance at a 0.05 level, the means were separated using the least significant difference test. Pearson's correlation analysis was used to quantify the relationships among initial litter chemistry, nutrient dynamics, and litter mass reduction. The analyses were conducted using IBM SPSS Statistics 23 and Microsoft Excel.

Results

Litter decomposition

The litter mass loss declined across the six species from the first to the fourth quarter (Fig. 2). Starting with an initial dry weight of 3 g, the first-quarter averages ranged from 1.88 g (*P. roxburghii*) to 2.60 g (*Quercus lanata*). By the fourth quarter, the weights further declined, with *P.*

Table 1 Selected tree species for the study and their location

Elevation (m)	Location	Coordinates	Aspect	Tree species	Types
1,029	Singati	27° 44' 23" N, 86° 11' 03" E	130° SE	<i>Pinus roxburghii</i> Sarg.	Evergreen
2,050	Laduk	27° 46' 40" N, 86° 10' 12" E	98° E	<i>Pinus wallichiana</i> A.B. Jacks.	Evergreen
2,156	Bulung	27° 46' 19" N, 86° 10' 40" E	137° SE	<i>Myrica esculenta</i> Buch.-Ham. ex D.Don	Deciduous
2,189	Bulung	27° 46' 19" N, 86° 10' 26" E	100° E	<i>Rhododendron arboreum</i> Sm.	Evergreen
2,243	Bulung	27° 46' 10" N, 86° 10' 23" E	34° SE	<i>Pinus patula</i> Schiede ex Schltdl. & Cham.	Evergreen
2,310	Bulungkhani	27° 47' 15" N, 86° 10' 10" E	144° SE	<i>Quercus lanata</i> Sm.	Evergreen

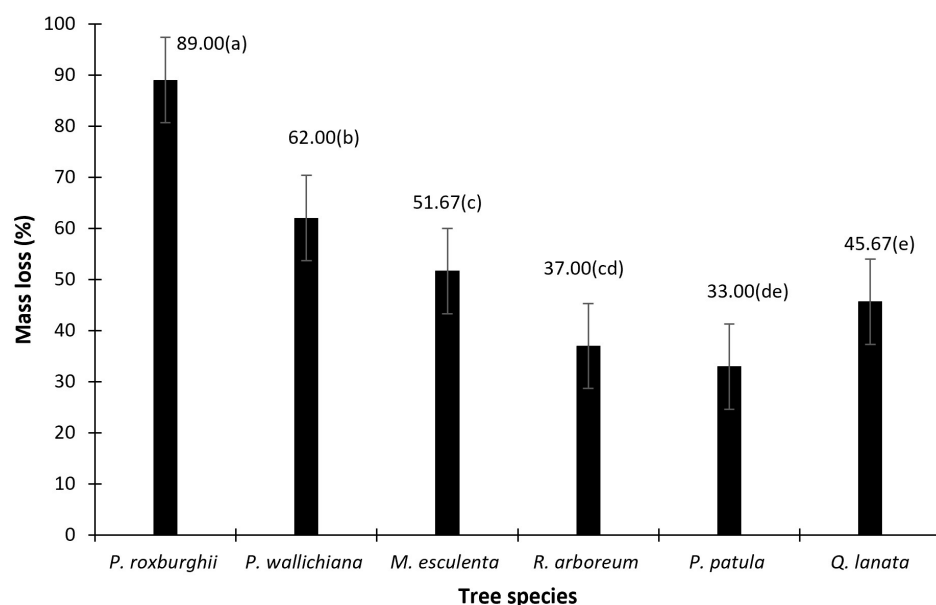


Fig. 2 Mean litter mass loss (%) of the six tree species after 12 months of decomposition. Error bars represent the standard error of mean of four replicates. Different lowercase letters indicate significant differences among tree species according to the least significant difference test at a 0.05 level of significance.

Table 2 Remaining litter mass (g of initial 3 g per litter bag) in a year-long *in-situ* litter decomposition

Species	First	Second	Third	Fourth
<i>Pinus roxburghii</i>	1.88 ± 0.05(a)	1.12 ± 0.09(a)	0.75 ± 0.15(a)	0.33 ± 0.05(a)
<i>Pinus wallichiana</i>	2.48 ± 0.01(c)	2.25 ± 0.09(c)	1.50 ± 0.07(b)	1.14 ± 0.10(b)
<i>Myrica esculenta</i>	2.18 ± 0.09(b)	1.89 ± 0.11(b)	1.52 ± 0.12(b)	1.45 ± 0.16(c)
<i>Rhododendron arboreum</i>	2.31 ± 0.09(b)	2.17 ± 0.08(c)	1.99 ± 0.06(c)	1.88 ± 0.12(de)
<i>Pinus patula</i>	2.67 ± 0.02(d)	2.52 ± 0.02(d)	2.35 ± 0.08(d)	2.01 ± 0.14(e)
<i>Quercus lanata</i>	2.62 ± 0.07(cd)	2.48 ± 0.01(d)	1.94 ± 0.08(c)	1.63 ± 0.14(cd)
F	93.33	177.90	124.26	94.96
p-value	< 0.001	< 0.001	< 0.001	< 0.001

Values are mean ± standard error of mean ($n = 4$), and lowercase letters denote significant differences at each sampling time within each column.

Table 3 Initial values of nutrients (mg g^{-1}) and their ratio among the tree species

Species	C _o	N _o	P _o	K _o	C:N	C:P	C:K	N:P	N:K	P:K
<i>Pinus roxburghii</i>	584.60	6.90	0.90	2.30	84.72	649.56	254.17	7.67	3.00	0.39
<i>Pinus wallichiana</i>	569.20	10.00	1.60	15.30	56.92	355.75	37.20	6.25	0.65	0.10
<i>Myrica esculenta</i>	567.20	20.00	0.50	3.20	28.36	1,134.40	177.25	40.00	6.25	0.16
<i>Rhododendron arboreum</i>	562.60	11.90	2.30	3.70	47.28	244.61	152.05	5.17	3.22	0.62
<i>Pinus patula</i>	576.70	18.50	0.70	6.30	31.17	823.86	91.54	26.43	2.94	0.11
<i>Quercus lanata</i>	562.10	10.00	1.70	5.50	56.21	330.65	102.20	5.88	1.82	0.31

C: carbon; N: nitrogen; P: phosphorus; K: potassium.

roxburghii showing the highest reduction (89%) and *Pinus patula* the lowest (33%).

Decomposition rates followed the order: *P. roxburghii* > *Pinus wallichiana* > *Myrica esculenta* > *Q. lanata* > *Rhododendron arboreum* > *P. patula*, ranging from 0.40 to 2.21. Half-life decay periods spanned 0.31–1.73 years, with 95% decay requiring 2.26–7.50 years (Table S1). Litter mass loss was highest at lower elevations, decreasing significantly with elevation (Table 2). Variations were evident between species and elevational zones (1,000–2,310 m).

Litter chemistry

The lowest N concentration was recorded in *P. roxburghii* at 1,010 m and the highest in *M. esculenta* at 2,156 m (Table 3). Similarly, the lowest P concentration was recorded in *M. esculenta*, and the highest in *R. arboreum*. Likewise, for K, the lowest concentration was recorded in *P. roxburghii* and the highest in *P. wallichiana*. The C concentration was highest in *P. roxburghii* and lowest in *Q. lanata*.

The differences in nutrient concentrations resulted in significant variations in nutrient ratios among the species. *Pinus roxburghii* had the highest C:N ratio while *M. esculenta*

lenta had the lowest. N:P ratio was the highest in *M. esculenta* and lowest in *R. arboreum*. Meanwhile, C:P and C:K were the highest in *M. esculenta* and *P. roxburghii*, respectively and the lowest in *R. arboreum* and *P. wallichiana*, respectively. *Myrica esculenta* and *R. arboreum* had maximum N:K and P:K ratios, respectively, whereas the minimum ratios of both were found in *P. wallichiana* (Table 3).

Litter nutrient dynamics

Pinus roxburghii consistently released the highest proportions of C and K across all time intervals, suggesting a relatively rapid decomposition rate for this species (Fig. 3). Conversely, the release patterns for N and P varied among species and time points, with different species exhibiting maximum release rates at different intervals. Additionally, the nutrient release patterns indicated significant variations among the tree species in terms of their ability to release specific nutrients from their decomposing litter over the course of a year (Fig. 3). *Pinus roxburghii* exhibited a nutrient release pattern of $P > C > K > N$, indicating a highest release of P (99.14) followed by C (96.81), K (95.69), and N (93.71). *Rhododendron arboreum* also showed a $P > C > K > N$ pattern with corresponding values of 97.27,

69.72, 69.43, and 67.26, respectively.

Similarly, *P. wallichiana* displayed a release pattern of $P > K > C > N$, with corresponding values of 99.29, 93.31, 83.38, and 82.55, respectively. However, for *P. patula*, the release pattern was $N > P > K > C$, with corresponding values 99.06, 95.22, 85.46, and 47.23, respectively. *Myrica esculenta* showed a release pattern of $P > N > C > K$, with values corresponding to 97.10, 76.79, 65.46, and 58.15, respectively. Finally, the release pattern of *Q. lanata* was $N > P > K > C$ with corresponding values of 99.51, 97.44, 82.21, and 50.96, respectively (Fig. 3).

Loss in litter mass had a strong correlation with nutrient release for all the species (Table 4). *Pinus roxburghii* showed near-perfect correlations with C ($r = 0.999$), N ($r = 0.998$), P ($r = 0.998$), and K ($r = 0.992$). *Pinus wallichiana* had very high correlations for C ($r = 0.995$), N ($r = 0.973$), and K ($r = 0.990$) but a lower correlation for P ($r = 0.766$). Similarly, *P. patula* exhibited strong correlations for all nutrients ($r = 0.998, 0.937, 0.935$, and 0.984 for C, N, P, and K, respectively). The broadleaf species also followed similar trends. *Rhododendron arboreum* had strong correlations with C ($r = 0.857$), N ($r = 0.914$), and P ($r = 0.870$), with K being the highest ($r = 0.996$). *Myrica esculenta* and *Q. la-*

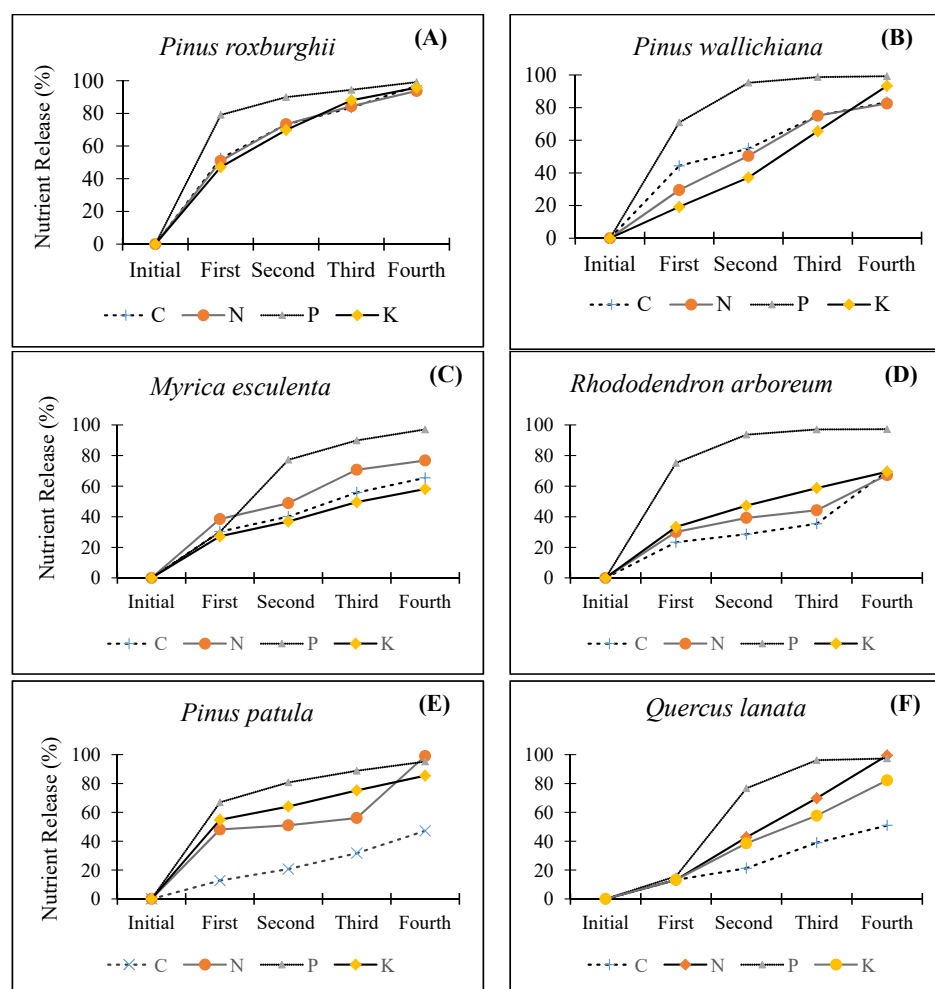


Fig. 3 Changes in carbon (C), nitrogen (N), phosphorus (P), and potassium (K) release (%) during one year of litter decomposition. Each panel (A–F) represents a different tree species and shows nutrient release from the initial sampling to the fourth quarter.

Table 4 Correlation between litter mass loss (%) and litter nutrient release (%)

Species	Litter C	Litter N	Litter P	Litter K
<i>Pinus roxburghii</i>	0.999**	0.998**	0.998**	0.992**
<i>Pinus wallichiana</i>	0.995**	0.973*	0.766	0.990**
<i>Myrica esculenta</i>	0.980*	0.991**	0.941*	0.981**
<i>Rhododendron arboreum</i>	0.857	0.914*	0.870	0.996**
<i>Pinus patula</i>	0.998**	0.937*	0.935*	0.984**
<i>Quercus lanata</i>	0.997**	0.975*	0.806	0.967*

C: carbon; N: nitrogen; P: phosphorus; K: potassium.

**Correlation is significant at the 0.01 level (1-tailed).

*Correlation is significant at the 0.05 level (1-tailed).

Table 5 Correlation between mass loss (%), initial litter contents (C_o , N_o , P_o and K_o in mg g^{-1}) and environmental parameters elevation (m) and aspect (0° to 360°)

	Mass loss	Elevation	Aspect	C_o	N_o	P_o	K_o
Mass loss	1						
Elevation	-0.911**	1					
Aspect	0.484	-0.262	1				
C_o	0.631	-0.800*	-0.265	1			
N_o	-0.613	0.563	-0.386	-0.164	1		
P_o	-0.230	0.247	0.115	-0.594	-0.542	1	
K_o	-0.023	0.270	-0.286	-0.147	-0.155	0.229	1

C: carbon; N: nitrogen; P: phosphorus; K: potassium.

**Correlation is significant at the 0.01 level (1-tailed).

*Correlation is significant at the 0.05 level (1-tailed).

lanata showed very high correlations across all nutrients, with C ($r = 0.98$ – 0.997) and N ($r = 0.975$ – 0.991) being the strongest.

Factors affecting litter decomposition

The first quarter (mid-June to mid-September) experienced high temperatures (maximum 23.75°C and minimum 16.75°C) and heavy rainfall (600 mm), favoring higher decomposition for *P. roxburghii*, *M. esculenta*, and *R. arboreum* (Table S2). The second quarter (mid-September to mid-December) saw moderate temperatures (maximum 19°C , minimum 10.75°C) and rainfall (262.5 mm), with lower decomposition rates, particularly for *P. wallichiana* and *R. arboreum*. The third quarter (mid-December to mid-March) was colder (maximum 14.75°C , minimum 5.5°C) and drier (150 mm of rainfall), where decomposition increased for *P. wallichiana* and *P. patula* but decreased for *P. roxburghii* and *R. arboreum*. The fourth quarter (mid-March to mid-June) experienced rising temperatures (maximum 20.67°C , minimum 10.67°C) and rainfall (383.33 mm), with higher decomposition for *P. patula* and *R. arboreum* but lower for *Q. lanata*.

The litter mass loss showed a significant negative correlation with elevation ($r = -0.911$, $p < 0.001$) and a moderate positive correlation with slope orientation (aspect, $r = 0.484$, $p < 0.001$), indicating microclimate influences on decomposition (Table 5). Litter nutrient concentrations also showed complex relationships: C_o negatively correlated with elevation ($r = -0.800$, $p < 0.001$) but positively with litter mass loss ($r = 0.631$, $p < 0.001$). Whereas N_o and P_o neg-

atively correlated with litter mass loss ($r = -0.613$ and $r = -0.230$, respectively, $p < 0.001$) (Table 5).

Discussion

Litter decomposition

This study showed that litter decomposition rates were more rapid at lower elevations than at higher elevations, consistent with Xiao et al. (2019). This may be because of variations in temperature, precipitation, and microbial interactions as reported by Semwal et al. (2003). However, *P. patula* decomposed slower than *Q. lanata* at similar elevations, highlighting the role of species-specific traits, such as litter chemistry, in overriding elevation effects. These results agree with Ghosh (2015), who found faster decomposition of *Quercus* litter in the Garhwal Himalaya, but is contrary to the results of Bohara et al. (2020); thus, emphasizing the context-specific nature of litter decomposition. Deciduous species like *M. esculenta* decomposed more rapidly than evergreens *R. arboreum*, aligning with established trends linking faster decay to shorter leaf lifespans in deciduous species (Cornwell et al. 2008). Decomposition rates followed the order: *P. roxburghii* > *P. wallichiana* > *M. esculenta* > *Q. lanata* > *R. arboreum* > *P. patula*, reflecting the influence of species on substrate quality and microenvironment (Eviner and Chapin 2003).

Rates of decomposition ranged from 0.40 to 2.21 yr^{-1} across elevational gradient (1,029–2,310 m), consistent with Kumar et al. (2016), who reported 0.26– 0.51 yr^{-1} at 2,050–

2,300 m in the temperate forest of the Garhwal Himalaya. Bohara et al. (2020) observed rates of 0.53–1.42 yr⁻¹ along an elevational gradient in central Nepal, overlapping with mid-range values. The higher or maximum rate (2.21 yr⁻¹) in this study likely reflects the inclusion of lower elevation sites with warmer, moister conditions.

Litter chemistry

Litter decomposition is influenced by its C, N, P, and K contents, with high-quality litter (high N, low C:N ratio) decomposing more rapidly, as reported by Gao et al. (2016), Zhang et al. (2018), and Xiao et al. (2019). Despite *P. roxburghii* having the highest C content (584.6) but low N (6.9) and P (0.9) and high C:N and C:P ratios, which are generally associated with slower decomposition, it decomposed faster at lower elevations. This suggests that warmer and humid conditions enhance microbial activity, as documented by Giweta (2020), or that other nutrient dynamics compensate for its low nutrient quality. In contrast, *P. patula*, with the high N content (18.5) and a low C:N ratio (31.17), decomposed slower at higher elevations. This is similar to the observation of Veen et al. (2015) which suggests that the cooler, drier conditions at these elevations limit microbial activity overriding the expected faster decomposition associated with lower C:N ratios. This is also consistent with the findings of Salah (2009), who observed increased decomposition rates for *P. patula* in warmer conditions in subtropical forest of Nagaland, North East India.

The findings also challenge the general threshold C:N ratio (20–30) for organic N mineralization, where microbial immobilization of N occurs (Bohara et al. 2020). However, *P. roxburghii* and *P. wallichiana* showed relatively rapid decomposition despite their high C ratios. This appears to be an exception in the present study and it may be associated with favourable temperature and moisture conditions, together with other factors such as microbial activity and litter characteristics (Aber and Melillo 1982; Rigobelo and Nahas 2004). Therefore, the observed pattern should be interpreted as the combined influence of litter quality and environmental conditions rather than temperature alone. These findings support previous research indicating that C:N ratios alone are not always reliable indicators of decomposition dynamics, as demonstrated by Vestgarden (2001) and Ghosh (2015).

Overall, the results suggest that both litter quality and climatic conditions influenced litter decomposition rates along the elevational gradient. Litter quality, particularly N concentration and C ratio, is generally considered as an important factor controlling decomposition (Cornwell et al. 2008; Gao et al. 2016). However, the contrasting behavior of *P. roxburghii* and *P. patula* indicates that favourable climatic conditions can modify the influence of litter quality. Although *P. roxburghii* had a high C ratio, it showed the fastest decomposition at lower elevations, whereas *P.*

patula decomposed slowly despite its relatively higher N content. This suggests that the warmer and wetter conditions at lower elevations enhanced microbial activity and accelerated litter breakdown, while the cooler conditions at higher elevations slowed the decomposition process (Aerts 1997; Berger et al. 2015).

Litter nutrient dynamics

Pinus roxburghii exhibited rapid and consistent nutrient release, aligning with its high decomposition rate. Despite the high lignin content in conifers like *Pinus*, decomposition at lower elevations was likely facilitated by lignin-degrading fungi, as suggested by other studies (Rigobelo and Nahas 2004). The substantial initial N and P release by *P. roxburghii* supports early plant growth and microbial activity, as observed by Gagnon and Zaidi (2021).

Similarly, *P. wallichiana* and *M. esculenta* exhibited gradual nutrient release, with K showing delayed release. This suggests the presence of tightly bound K forms require prolonged microbial activity, as described by Laskowski et al. (1995). In contrast, Han et al. (2005) observed rapid initial K release in *Malus domestica*, highlighting species-specific variations in nutrient dynamics. The steady nutrient release in *P. wallichiana* is consistent with findings of Laskowski et al. (1995), indicating its role in maintaining long-term ecosystem stability by ensuring continuous nutrient availability.

Pinus patula exhibited slow C release but rapid N and P release, suggesting that while limited initial C release may constrain microbial growth, it could contribute to long-term soil OM accumulation as reported by Jat et al. (2022). *Rhododendron arboreum* showed rapid P release in the early stages, a trend also observed by Bisht et al. (2014) in temperate forests of the Central Himalayas, benefiting P-deficient soils and promoting seedling growth as reported by Qusted and Eriksson (2006). The balanced release of other nutrients in *R. arboreum* aligns with the findings of Dong et al. (2022), who noted that sustained microbial activity is promoted by consistent nutrient availability.

In the case of *Q. lanata*, the slow initial nutrient release suggests the presence of recalcitrant litter components, a trend previously reported by Murphy et al. (1998). The overall pattern of high P release across species aligns with observations in the Central Himalayas (Bohara et al. 2020; Saha et al. 2016). However, *P. patula* and *Q. lanata* showed the highest N release at the end of decomposition, which may be attributed to their high initial N content or the role of decomposers such as soil microarthropods as reported by Giweta (2020).

The slow N release observed in most species is consistent with the findings of Bohara et al. (2020), who suggested that there is strong N retention in litter layer. Variations in nutrient dynamics during decomposition also align with the findings of Singh et al. (2007), highlighting the influence of climatic and phenological factors, as further sup-

ported by the studies of Saha et al. (2016). Additionally, the positive correlation between litter mass loss and nutrient release in this study contrasts with Bohara et al. (2020), who reported negative correlations except for N. This discrepancy is likely due to differences in plant traits, climate, temperature, moisture, and microbial activity, as discussed by Bhattarai and Bhatta (2020) and Du et al. (2020).

Factors affecting litter decomposition

During the first quarter, higher temperatures and rainfall significantly accelerated decomposition, particularly for *P. roxburghii* and *M. esculenta*, aligning with the findings of Li et al. (2022). In contrast, the second quarter's moderate climatic conditions appeared to slow decomposition, likely due to cooling temperatures and reduced precipitation, as supported by Giweta (2020). The lower temperatures and reduced rainfall in the third quarter, most likely, suppressed microbial activity, leading to an overall decrease in decomposition rates, a pattern also observed by Archer (2004).

Similar to the findings of Salah (2009), the decomposition rates of *P. wallichiana* and *P. patula* were relatively higher, likely due to specific litter traits or localized microclimatic factors. With increasing temperatures and rainfall in the fourth quarter, *P. patula* and *R. arboreum* exhibited accelerated decomposition, consistent with the findings of Singh et al. (2007) and Saha et al. (2016), which emphasize the role of moisture in enhancing litter breakdown.

A strong negative correlation between litter mass loss and elevation is in agreement with the findings of Bohara et al. (2020), indicating that decomposition rates are reduced at higher elevations due to colder, drier conditions and shifts in microbial communities. Additionally, the observed correlations between initial litter nutrient content and stoichiometric ratios highlight the role of nutrient interactions in regulating mobility during decomposition as reported by Bohara et al. (2020).

The present findings indicate that climatic conditions and litter quality acted together in regulating decomposition. Temperature and moisture associated with elevation influenced microbial activity, while differences in litter nutrient concentrations and C ratios affected the rate of litter breakdown among species (Cornwell et al. 2008; Taylor et al. 1989). Therefore, the observed variation in decomposition rates cannot be explained by climate or litter chemistry alone but is likely due to their combined influence. Overall, the observed links between litter stoichiometry, breakdown rates and nutrient composition align with the conclusions of Taylor et al. (1989), reinforcing the influence of litter chemistry and physical traits on decomposer activity. Further long-term studies are required to evaluate the later stages of litter decomposition and nutrient dynamics, particularly for slowly decomposing species at higher elevations.

Conclusions

Our study demonstrated that variation in litter decomposition among the Himalayan tree species was driven by the combined effect of litter chemistry and topographic conditions. Decomposition and nutrient release rates were higher at lower elevations despite high C:N ratios. This indicated that favorable microclimatic conditions and enhanced microbial activity promoted more rapid litter decomposition. Overall leaf litter decomposition rate was in the order: *P. roxburghii* > *P. wallichiana* > *M. esculenta* > *Q. lanata* > *R. arboreum* > *P. patula*. The estimated time required for complete decomposition was approximately five times longer for *P. patula* (about 13 years) than for *P. roxburghii*. This highlighted great variation in litter decomposition rates among closely related species across different elevation and suggested that slower decomposition at higher elevations contributes to the accumulation of OM in the Himalayan soils.

Since this study was conducted over a year and only represented the initial stage of litter decomposition, future research should be focused on long-term monitoring to observe the complete decomposition process. Studies with larger sample size with continuous *in situ* microclimatic measurements using data loggers will enhance the understanding of later stages of decomposition, nutrient dynamics, and the effect of ongoing climate change. Furthermore, integrating the influence of anthropogenic disturbances with climatic drivers will provide a more comprehensive understanding of nutrient cycling and decomposition patterns in the Himalayan forest ecosystems under changing environmental conditions.

Supplementary Information

Supplementary information accompanies this paper at <https://doi.org/10.5141/jee.26.029>.

Table S1. Litter decay rate (K), half-life (t50), and the time required for 95% decay (t95) from the residual litter of different forest types. **Table S2.** Litter decomposition (mass loss in %) of each quarter in a year-long *in situ* litter decomposition process.

Abbreviations

OM:	Organic matter
C:N ratio:	Carbon-to-nitrogen ratio
N:	Nitrogen
C:	Carbon
P:	Phosphorus
K:	Potassium
GCA:	Gaurishankar Conservation Area
ANOVA:	Analysis of variance

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Authors' contributions

SA: Conceptualization, Investigation, Formal analysis, Writing - original draft; MPD: Conceptualization, Funding acquisition, Methodology, Visualization, Investigation, Writing - review & editing, Supervision. KPS: Conceptualization, Methodology, Visualization, Writing - review & editing, Supervision.

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Availability of data and materials

The datasets generated during this study are available from the corresponding author upon reasonable request.

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare that they have no competing interests.

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